
C Programming Language – General Study Summary

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1. Introduction to C

- **C** is a **procedural programming language**.
- Known for **performance, portability**, and **low-level memory access**.
- Used in **system programming**, **embedded systems**, **operating systems**, etc.

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2. Basic Structure of a C Program

```
#include <stdio.h>   // Preprocessor directive


int main() {        // Main function – entry point
   printf("Hello, World!");
   return 0;       // Exit code
}
```

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3. Data Types

Type	Description	Example
int	Integer	int a = 5;
float	Floating point number	float b = 3.5;
char	Single character	char c = 'A';
double	Double precision float	double d = 2.3456;

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4. Operators

- **Arithmetic**: +, -, *, /, %
- **Relational**: ==, !=, <, >, <=, >=
- **Logical**: &&, ||, !
- **Assignment**: =, +=, -=, etc.
- **Increment/Decrement**: ++, --
- **Compound Operator:** a += 5 means a = a + 5
- **Increment Operator:** ++a increments a before it's used.
- **Integer Division:** int a = 19 / 8; result is 2, not 2.375.
- **To get floating-point division:** One of the operands must be a float.

5. Comments

- **Single-line:** // This is a comment
- **Multi-line:** /* This is a multi-line comment */

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6. Control Structures

- **if / else if / else** – Conditional branching
- **switch** – Multi-way branching
- **for loop** – Definite iteration
- **while loop** – Conditional iteration
- **do-while loop** – Executes at least once
- **break** – Exits loops or switch
- **continue** – Skips current loop iteration

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7. Functions and Keywords

- **Function Exit:** Use return to exit a function and optionally return a value.
- **Function Calls:** Functions can be called **unlimited times** in a program.

- **Decision-Making Keyword:** if, used with else and else if for conditional logic.
- **Loop Exit:** Use break to exit from loops prematurely.
- **Skip Iteration:** Use continue to skip the current iteration of a loop.

8. Loops

- **Types of Loops:** for, while, do-while
- **Guaranteed to run once:** do-while
- **break:** Exits the loop immediately.

continue: Skips the rest of the loop body and goes to the next iteration.

9. Input/Output

```
// Input
scanf("%d", &number);


// Output
printf("Value is: %d", number);
```

10. Format Specifier Meaning

%d	Integer
%f	Float
%c	Character
%s	String

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11. Functions

- Block of reusable code.

```
int add(int a, int b) {
   return a + b;
```

}

- **Main function:** int main() is required in every program.
- **Return types:** int, void, etc.

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12. Arrays

- **Declaration:** int arr[5];
- **Initialization:** int arr[] = {1, 2, 3};
- **2D Arrays:** int matrix[3][3];
- Arrays use **0-based index**
- **1D Array Declaration:** int arr[3];
- **2D Array Declaration:** int arr[3][3];
- **Accessing Elements:** arr[2] gives the 3rd element (0-indexed).
- **Out-of-Bounds Access:** Accessing arr[3] in int arr[3] = {10, 20, 30}; leads to **undefined behavior**.

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13. Strings

- Strings are arrays of characters ending with \0.

char name[] = "John";

- Use printf("%s", name); and scanf("%s", name); for input/output.

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14. Comments

- **Single-line:** // This is a comment
- **Multi-line:** /* This is a multi-line comment */

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15. Common Errors in C

- Missing semicolon ;
- Unmatched braces {} or parentheses ()
- Forgetting & in scanf()
- Buffer overflow with arrays
- Using undeclared variables

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16. Advantages of C

- Fast and efficient
- Portable across platforms
- Close to hardware (low-level)
- Foundation for many modern languages (C++, Java, etc.)